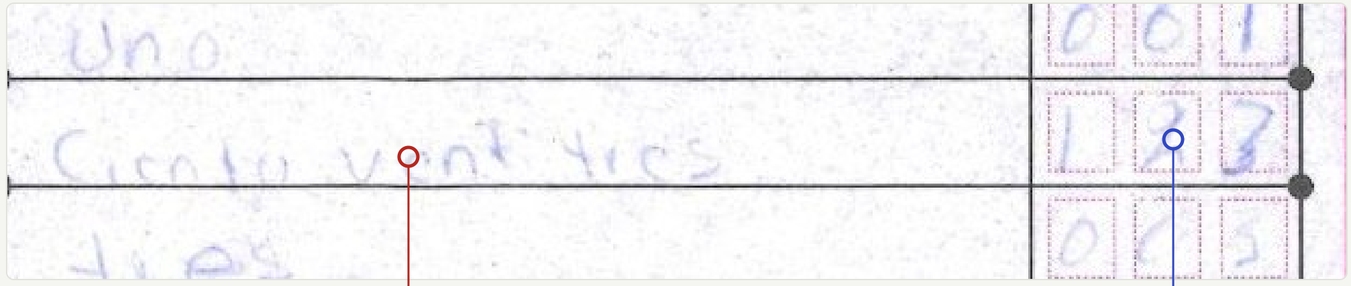
# Two Hands

## Verifiable quality control for handwritten numbers.

OCR reads the number. Two Hands checks whether the whole form agrees.



**Gemma read 23**
in digits and in words: rows sum to 366, TOTAL is 466

**The paper: 123**
routed to a person, with this crop

Form 122655C0100, Option 02. INE 2024 tally sheet, privacy-masked, cropped right of the logo column.

---

**1,270**
real handwritten tally sheets read by Gemma on AMD

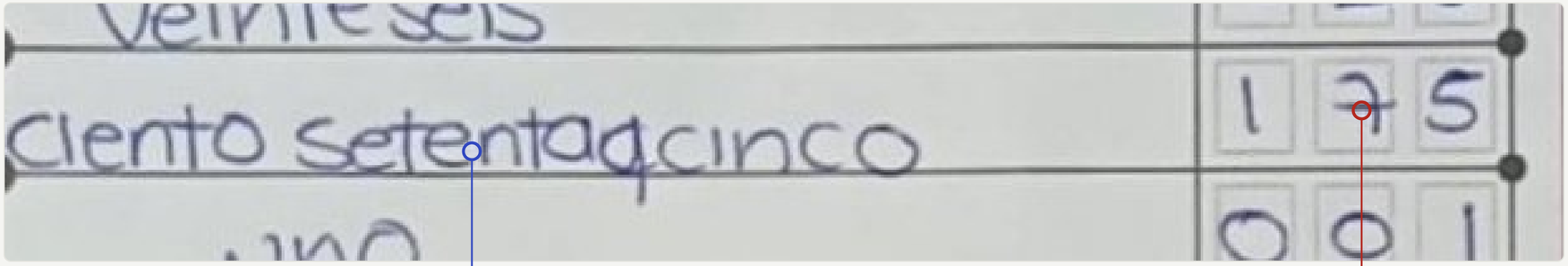
**91.6%**
field match, vs 62.84% for the reading the state bought

**27** of 28
certified-detectable capture errors caught

**$1**
per sheet audited (proposed)

THE PROBLEM, FROM INE'S OWN FILES

# Two official records of the same paper disagree. Neither says which one is right.




The paper: 175
in words and in boxes; the certified count says 175

First capture keyed 75
the preliminary record

**6.5%**

of the 52,167 polling stations not recounted: the preliminary capture and the certified count disagree on at least one vote field

**19.2%**

of nationally captured tally forms fail their own row sum (26,336 of 137,196)

Double keying copies the paper faithfully but **cannot say which paper is wrong**. The photo can.

HOW IT WORKS

# The model reads. The paper's own arithmetic decides.

01

## Photo

Page located and privacy-masked before it is stored or sent

→

02

## Gemma reads

Gemma-4-31B-IT on AMD Token Factory reads every field twice: digits and words

→

03

## Arithmetic

words = digits · rows sum to TOTAL · section 5 = section 7 = TOTAL

→

04

## Accept or route

All hold: **accepted**. One fails: **a person** gets the field, with its photo

05

## Receipt

Each field a Merkle leaf; the root anchored with OpenTimestamps

A second reader, Qwen3.8-27B on AMD, never overrides Gemma: it can only **withhold** acceptance.

ONE REAL SHEET, REQUEST TO RECEIPT

# Gemma read 23. The paper says 123. The sum caught it.

| IDENTITY | LEFT SIDE | RIGHT SIDE | RESULT |
|---|---|---|---|
| **rows sum to TOTAL** | **366**<br>Option 01 1 + Option 02 23 + Option 03 3 + Option 04 26 + Option 05 22 + Option 06 98 + Option 07 165 + Option 08 0 + Option 09 0 + Option 10 0 + Option 11 0 + Option 12 7 + Option 13 0 + Option 14 0 + Option 15 0 + Unregistered candidates 0 + Null votes 21 | **466**<br>TOTAL 466 | FAIL |
| **votaron + representantes = section 5** | **466**<br>Voters on the list 466 + Representatives who voted 0 | **466**<br>Voters + representatives 466 | pass |
| **section 5 = section 7** | **466**<br>Voters + representatives 466 | **466**<br>Ballots taken from the boxes 466 | pass |
| **section 7 = TOTAL** | **466**<br>Ballots taken from the boxes 466 | **466**<br>TOTAL 466 | pass |

/trace on the site: the gate re-run on Gemma's stored raw output for form 122655C0100.

THE READ

23 in digits and 23 in words: they agree, so only the arithmetic can catch it.

THE GATE

Rows sum to 366, the form's TOTAL is **466**: routed to a person. With the paper's 123: 466 = 466.

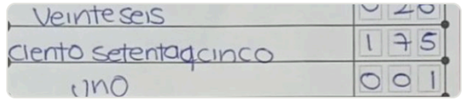
THE RECEIPT

VERIFIED: the record is unchanged since the report was sealed. Not that the reading is right.

RESULTS · PRE-REGISTERED · INE'S PUBLIC DATA

# 1,270 real tally sheets read on AMD. 49 capture errors confirmed by the certified count.

3




In English: words: "ciento setenta y cinco" = 175 · boxes: 1 7 5

**050815B0100 · Option 07** STRONG

receipt

Judge this verdict yourself

| Keyed | Model digits | Model words |
|---|---|---|
| 75 | 175 | 175 |

- model digits and words agree (175)
- the keyed value breaks rows sum to TOTAL (258 vs 358); the model's value satisfies it

One dispute in the audit report: the crop, the keyed 75, Gemma's 175 in digits and words.

### RANDOM 300, PRE-REGISTERED

**52** fields where the official capture does not match the paper, in 6,358 checked; each with its photo.

### COLIMA, A WHOLE STATE

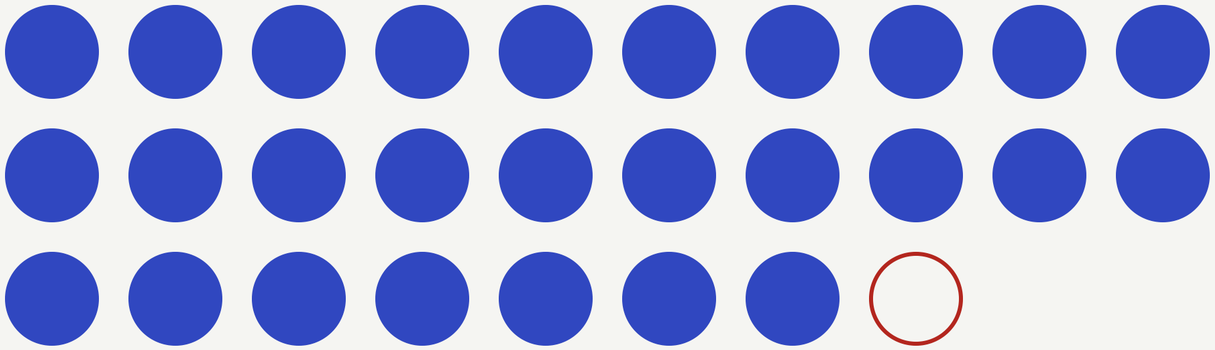
All **982** located presidential sheets for **$0.14**; 27 fields where the capture does not match the paper, each confirmed by the certified count.

### 100 FRESH FORMS

Gate v2 accepted 22 forms: **0 of 484** fields wrong.

CHECKED

# 27 of 28 capture errors caught. A blind re-read found 0 differences.



Each dot: a capture error the certified count reveals in Colima. Filled: caught. Outlined: missed.

RECALL

**27 of 28**, 96% (Wilson 95% 82-99%). One state, pre-registered; certified-detectable errors only, not total recall.

A PERSON'S BLIND RE-READ

of the AI-judged disputes: **0 read differently**.

AGAINST THE INCUMBENT · BAJA CALIFORNIA SUR, 2024 LOCAL ACTAS

# 91.6% field match, where the reading the state bought scored 62.84%.

**Two Hands**
Gemma on AMD · random 300 actas, pre-registered



91.6%

**Google Document AI**
the state's purchased reading · all 2,020 actas



62.84%

Same per-field exact-match metric. Ours: 3,622 fields, Wilson 95% 90.63-92.44%, against the published PREP values. Theirs: 14,501 of 23,077 fields, against the manual capture.

GEMMA ON AMD TOKEN FACTORY

# Same 300 forms, same gate: Gemma accepts 1.8× the forms Qwen does, 4.6× cheaper, 2.8× faster.

| READER, ON AMD | FORMS ACCEPTED | WRONG OF ACCEPTED FIELDS | $ PER 1,000 FORMS | TIME PER READ |
|---|---|---|---|---|
| **Gemma-4-31B-IT** | **55 of 300** | **2 of 1,210** | **$0.14** | **87 s** |
| Qwen3.8-27B | 31 of 300 | 1 of 682 | $0.64 | 241 s |
| Both must agree, the paper's sums as referee | 89 of 300 | **0 of 1,958** | $0.78 | – |

Two readers: lighter gate, a secondary analysis. Qwen: +2 undecided. Source: app/data/bakeoff_300.json.

**4.8** forms / min

$0.000146 per form, shared endpoint, 4 in flight

**$0.14**

Colima, a whole state: all 982 located presidential sheets

**150** of 150

Evolus "Audit finding" runs completed; the agent runs Gemma on AMD

BUSINESS AT A GLANCE

# An independent check of the keyed values. $1 per sheet audited (proposed).

TWO HANDS · AUDIT OF A KEYED CAPTURE AGAINST ITS PAPER

## Audit report: 6,358 fields checked, 267 disputed, each with its photo

PREP 2024, 300 pre-registered forms · 300 forms in the keyed file, 289 compared · audit `ab2b0555076b1` · 2026-10-17 · Merkle root `e56a8ef4f011b9e5` · fingerprint `c4788f2501814fd1`

### Summary

| | |
|---|---|
| Fields checked (keyed value vs the model's read of the paper) | 6,358 |
| Disputed fields (the paper, as read, disagrees with the keyed value) | 267 |
| with strong evidence that the keyed value is wrong | 18 |
| with medium evidence | 80 |
| with weak evidence (the model may have misread) | 169 |
| Marked by the auditor: confirmed capture errors / not an error | 0 / 0 |
| Forms not compared (no read of the paper yet, or page not located: a person checks them) | 11 |

The product: the audit report, every dispute with its photo, sealed by one Merkle root.

BUYER

INE and the 32 state institutes, each with an outside auditor.

WHY NOW

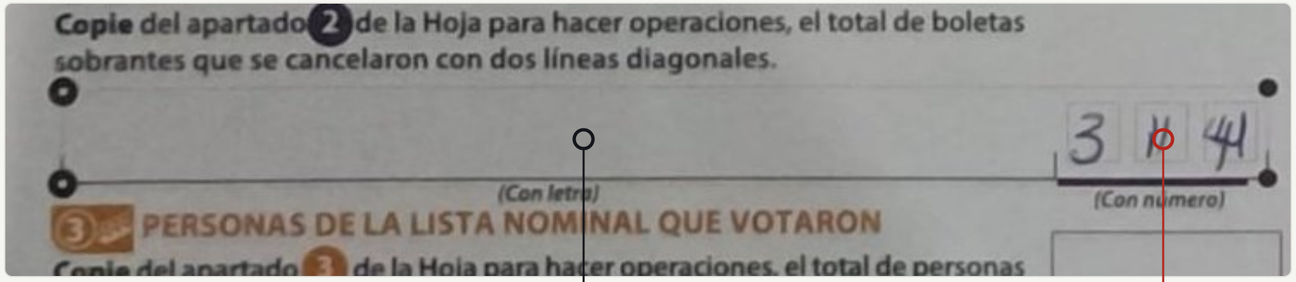
**6 June 2027**: the federal midterm and 32 concurrent local elections. First step: a **paid shadow-mode pilot** on one state's count.

ONE STATE, PRICED

Morelos paid MXN 26.6M to run its 2024 PREP. Our check of every sheet: about **MXN 143k**, 0.5% of that.

WHERE IT HOLDS, AND WHERE IT BREAKS

# Read 3114, paper 314. Gate v2 sends it to a person.




Words line: blank
no second reading to compare

The model read 3114
the paper says 314

Form 070212B0100, unused ballots (masked crop).

THE BLIND SPOT

Unused ballots sit in no identity: no sum can catch a digit inserted here.

HELD OUT

On 100 fresh forms the frozen gate let it through. Gate v2, digits and words both read and equal, blocked it: **0 wrong of 484**.

SCOPE

Clerical capture quality only. Most forms still go to a person. Not an official INE product.

CHECK IT YOURSELF

# Every number re-derives from public data.

REPRODUCE

```
make eval
```

Recomputes every number offline and checks it against what the site shows.

VERIFY A RECEIPT

```
python -m twohands.verify_receipt
  receipt.json --crop crop.jpg
```

A changed value or crop fails.

CHECK THE TIMESTAMP

```
ots verify
results/sample_audit_root.txt.ots
```

The report's Merkle root, anchored in Bitcoin with OpenTimestamps.

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**Two Hands.** **Verifiable quality control for handwritten numbers.** `github.com/py-tr/twohands`